KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

YEAR 7 MATHEMATICS · CHAPTER 11 PROBABILITY

Complementary events

WORKSHEET · WORK ON YOUR OWN

MA4-PRO-C-01
Solves problems involving the probabilities of simple chance experiments

KEY WORDS

outcome	one thing that can happen. A die has 6: 1, 2, 3, 4, 5, 6.
event	what we want to happen, e.g. rolling an even number.
complement	all the **other** outcomes. It is the event **not** happening.
not	P(not A) means "the chance that A does **not** happen".

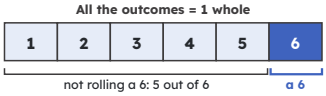
How to use this sheet: read each grey **Example**, then do the blue **Round** under it. Check your answers at the end of page 2.

Notes: An event and its complement never share an outcome. Together they make **all** the outcomes, so one of them **must** happen.

FORMULAS

$$P(\text{not A}) = 1 - P(A)$$

$$P(A) + P(\text{not A}) = 1$$



Tip: 1 whole = $\frac{4}{4} = \frac{6}{6} = \frac{10}{10} = 1.0$

EXAMPLE 1 Name the complement and list its outcomes

Use me for **Round 1** ↓

A die is rolled. **Event:** rolling a 2. What is the complement? List its outcomes.

1. List **all** the outcomes. Cross out the event (rolling a 2): 1 ~~2~~ 3 4 5 6
2. What is left is the complement. **Complement: not rolling a 2. Outcomes: 1, 3, 4, 5, 6**

ROUND 1 Easy Support

↑ Use Example 1 · fill in the blanks

	What happens	Complement (in words)	Outcomes in the complement
1	A die is rolled. **Event:** rolling a 5	not rolling a ____	1, 2, 3, 4, ____
2	A die is rolled. **Event:** an even number (2, 4, 6)	rolling an _____ number	____, ____, ____
3	Spin this spinner. **Event:** landing on red (R) 	not landing on _____	_____, _____, _____
4	Red, green and orange lollies in a bag. **Event:** picking green	not picking _____	_____ or _____
5	A coin is tossed. **Event:** getting heads	_____	_____

EXAMPLE 2 Find P(not A)

Use me for **Round 2** ↓

A die is rolled. P(rolling a 6) = $\frac{1}{6}$. Find P(**not** rolling a 6).

1. Use the rule: P(not 6) = 1 − P(6). Write 1 with the **same bottom number**: $1 = \frac{6}{6}$
2. Take away the tops: $\frac{6}{6} - \frac{1}{6} = \boxed{\frac{5}{6}}$ Check: not 6 is 1, 2, 3, 4, 5. That is 5 out of 6 ✓

ROUND 2 Easy Support / Core

↑ Use Example 2 · fill in the blanks

6 P(A) = $\frac{1}{4}$. Find P(not A).

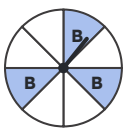
P(not A) = $1 - \frac{1}{4} = \frac{4}{4} - \frac{1}{4} = \frac{\square}{\square}$

7 The chance of rain is $\frac{2}{5}$. Find P(no rain).

P(no rain) = $1 - \frac{2}{5} = \frac{___}{5} - \frac{2}{5} = \frac{\square}{\square}$

8 3 of the 8 equal parts are blue (B).
P(blue) = $\frac{3}{8}$. Find P(not blue).

P(not blue) = $1 - \frac{\square}{\square} = \frac{\square}{\square}$



9 P(the bus is late) = $\frac{1}{10}$.
Find P(the bus is **not** late).

P(not late) = $1 - \frac{\square}{\square} = \frac{\square}{\square}$

10 **On your own.** A bag has 7 marbles. 2 are red.
P(red) = $\frac{2}{7}$. Find P(not red).

Working

Answer: _____

EXAMPLE 3 Check that P(A) + P(not A) = 1

Use me for Round 3 ↓

a Lily: P(win) = $\frac{3}{8}$, P(not win) = $\frac{5}{8}$. Right?

(1) Add them: $\frac{3}{8} + \frac{5}{8} = \frac{8}{8}$

(2) $\frac{8}{8}$ = 1 whole. **Yes, Lily is right ✓**

b Sam: P(snow) = $\frac{1}{3}$, P(no snow) = $\frac{1}{3}$. Right?

(1) Add them: $\frac{1}{3} + \frac{1}{3} = \frac{2}{3}$

(2) $\frac{2}{3}$ is not 1. **No ×. P(no snow) = 1 − $\frac{1}{3}$ = $\frac{2}{3}$**

ROUND 3 Medium Core

↑ Use Example 3

11 Add: $\frac{2}{7} + \frac{5}{7} = \frac{\square}{\square} = ____$

Do they add to 1? Yes No

12 P(A) = $\frac{4}{9}$ and P(not A) = $\frac{5}{9}$.

Add: $\frac{\square}{\square} + \frac{\square}{\square} = \frac{\square}{\square}$

Do they add to 1? Yes No

13 Ben says P(win) = $\frac{1}{6}$ and P(not win) = $\frac{4}{6}$.

Do they add to 1? If not, what should P(not win) be?

Working

Answer: _____

14 Mia says P(goal) = $\frac{3}{10}$ and P(no goal) = $\frac{7}{10}$.

Is Mia right? Show how you know.

Working

Answer: _____

EXAMPLE 4 Decimals and everyday problems

Use me for Round 4 ↓

a The chance of rain is 0.3. Find the chance of **no** rain.

(1) 1 whole = 10 tenths. 0.3 = 3 tenths.

(2) 10 − 3 = 7 tenths. **1 − 0.3 = 0.7**

b 12 pencils. 5 are red. One is picked. Find P(not red).

(1) Not red: 12 − 5 = 7 pencils

(2) 7 out of 12. **P(not red) = $\frac{7}{12}$**

ROUND 4 Medium Core

↑ Use Example 4

15 The chance Jack wins is 0.4.
Find the chance he does **not** win.

1 − 0.4 = _____

16 9 lollies. 4 are orange. One is picked.
How many are **not** orange? ____

P(not orange) = $\frac{\square}{\square}$

17 A die is rolled.
P(1 or 2) = $\frac{2}{6}$.
Find P(**not** 1 or 2).

Working

Answer: _____

18 3 of the 11 students in a team wear glasses. One is picked. Find P(**not** glasses).

Working

Answer: _____

CHALLENGE Challenging Core (stretch)

↑ Use all the examples

19 12 equal parts: 5 red, 4 blue, the rest yellow. Find P(**not** yellow).



Working

Answer: _____

20 10 marbles, each red or blue.
P(**not** red) = $\frac{3}{10}$.
How many are red?

Working

Answer: _____

ANSWERS **1** 5; 1, 2, 3, 4, 6 **2** odd; 1, 3, 5 **3** red; blue, green, yellow **4** green; red or orange **5** getting tails; tails **6** 3/4 **7** 5; 3/5 **8** 3/8; 5/8 **9** 1/10; 9/10 **10** 5/7 **11** 7/7 = 1; Yes **12** 4/9 + 5/9 = 9/9 = 1; Yes **13** 1/6 + 4/6 = 5/6, not 1; P(not win) = 5/6 **14** Yes: 3/10 + 7/10 = 10/10 = 1 **15** 0.6 **16** 5; 5/9 **17** 4/6 (or 2/3) **18** 8/11 **19** 3 yellow; 9/12 (or 3/4) **20** P(red) = 7/10, so 7 red